

Extreme Programming Test Case Description

Key Components of a Test Case in XP:

1. **Test Case ID:** A unique identifier (e.g., TC-001).
2. **Title/Name:** A brief description of what the test is verifying.
3. **Related User Story:** Links the test to a specific user story.
4. **Preconditions:** Any setup required before executing the test.
5. **Test Data:** Input values needed for the test.
6. **Test Steps:** Step-by-step instructions to perform the test.
7. **Expected Result:** The desired outcome if the feature works correctly.
8. **Actual Result:** Filled in after execution to confirm if it matches the expected result.
9. **Status:** Pass/Fail result after execution.

Example: Test Case for Sustainable Shopping Assistant App

User Story:

"As a shopper, I want to scan a product's barcode so that I can instantly see its sustainability rating."

Test Case Description

- **Test Case ID:** TC-001
- **Title:** Verify Barcode Scanning Displays Correct Sustainability Rating
- **Related User Story:** US-001 (Barcode Scanning Feature)
- **Preconditions:**
 - The app is installed and running.
 - The user is logged in.
 - Barcode scanner permissions are granted.
- **Test Data:**
 - Product Barcode: 1234567890123
 - Expected Sustainability Rating: **"A+" (Eco-Friendly)**
- **Test Steps:**
 1. Open the Sustainable Shopping Assistant App.
 2. Navigate to the barcode scanning feature.
 3. Point the camera at the barcode 1234567890123.

4. Wait for the app to process the barcode.
 - **Expected Result:**
 - The app successfully scans the barcode.
 - The correct product name is displayed.
 - The sustainability rating “**A+**” (**Eco-Friendly**) is shown clearly.
 - **Actual Result:** *(To be filled after testing)*
 - Example: “Barcode scanned successfully, rating displayed as A+.”
 - **Status:** *(Pass/Fail after execution)*
 - Example: ☒ **Pass**
-

Additional Test Cases for the Same Feature:

1. **TC-002:** Validate error message for an invalid barcode.
 2. **TC-003:** Verify app behavior when the camera permission is denied.
 3. **TC-004:** Check the barcode scanning feature’s performance under poor lighting conditions.
-

Key XP Testing Principles:

- **Automated Tests:** In XP, these test cases are often automated using unit testing frameworks (e.g., JUnit for Java, XCTest for iOS, etc.).
- **Continuous Feedback:** Tests are run frequently to catch issues early.
- **Simple and Clear:** Tests focus on clear, concise validation of functionality.